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Argumentation

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The goal of a scientific report is to convince your reader of the conclusions of your research project. In this section, you will learn more about the tools you can use for that aim: argumentation, proper word choices, use of references, and figures and tables. This first section will deal with argumentation.

Argumentation

The most important tool to convince your readers of your result, is argumentation: you show the chain of reasoning that leads from your assumptions to your conclusions. Each statement in this chain is either introduced as an assumption, or supported by evidence (a calculation perhaps, or a reference), or follows logically from the preceding statements.

To get a feeling for this, it might help to embrace your inner three-year-old: Ask yourself "why?" after every single sentence. Each of these questions should be answered in the report itself to provide a complete chain of reasoning.

Exercise A

1/1 point (graded)

4.3.3: Determining the values of diffusion constants

The three derivatives in our model depend on unknown constants d_1 , d_2 and d_3 . We have estimated the values of these constants in the following way:

d_1 : the diffusion rate of the medicine from the stomach into the bloodstream. If the concentration of the medicine in the stomach exceeds the concentration in the bloodstream, the medicine will diffuse at rate d_1 . If the value of d_1 is low, the diffusion process is slow. For a higher value of d_1 , the diffusion will be faster. We have chosen a low value of d_1 , as our setting concerns a pregnant woman and multiple processes are occurring in her body. We picked the value $d_1 = 0.2$, since this value produced the smoothest graph.

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The text above is part of a mathematical modelling report. The chain of reasoning in this paragraph is flawed. Which question(s) are not answered satisfactorily?

☐ Why do we need to estimate the constants d_1 , d_2 and d_3 ?

☒ Why have the authors chosen a low value of d_1 ? ✓

☒ Why did the authors pick $d_1 = 0.2$? ✓

Answer

Correct:

selected: The authors provide the explanation "our setting concerns a pregnant woman and multiple processes are occurring in her body", but it is not clear why this implies a low value of d_1

Submit

i Answers are displayed within the problem

Connecting your statements

To clarify your argumentation, it helps to make clear how your sentences (or parts of sentences) are related to each other. For example, in the sentence

"We have picked the value $d = 2$, since this value produced the smoothest graph."

the second half of the sentence provides a motivation for the first half. The word "since" connecting the two clauses of this sentence, shows the relation between them.

Connections between clauses, sentences or paragraphs can help the reader understand the flow of the argument presented in the text. They turn a collection of sentences into a coherent unit, which is easier to read. Let us take a look at some of the different connections you might want to make:

- **Causality:** One sentence / clause / paragraph is a consequence of another one. Common ways to connect these components are words and phrases like: because, since, as, for, therefore, thus, as a result, so, hence. There are many more options to choose from. This allows you to avoid using the same word over and over again, making your text more pleasant to read.

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despite, compared to, whereas, in contrast to, yet.

There are many other possible connections between clauses, sentences or paragraphs. What connections are implied, for example, by the following words?

And, or, nor, both ... and ... , rather ... than ..., not only ... but also ..., unless, whenever, whether ... or ..., if, in order to

If you have trouble finding convenient words to express your ideas, a thesaurus can help you find the words you are looking for. This can help you write a varied and coherent text, which is easy to understand.

Note of warning: It is not necessary to use connecting words for every single sentence.

Exercise B

1/1 point (graded)

 Keyboard Help

PROBLEM

The following simplified sentences all concern causality, so either A is a consequence of B or vice versa. Drag and drop the sentences to the field that describes the correct causality.

Because A, B. A, so B. A gives rise to B. A, since B. A, owing to B. A, as B.

As a result of A, B. A. Therefore B. The reason for A is B.

A, thus B. A, thereby B. A, as a consequence of B.

One consequence of A is B.

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You have used 2 of 10 attempts.

 Show Answer

✓ Your highest score is 1.0

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